Preprocessing Report: Stellar Classification

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Introduction

- The goal of this project is to classify astronomical objects. Specifically stars, galaxies, and quasars using spectral data from the Sloan Digital Sky Survey. The dataset contains 100,000 observations with multiple photometric, positional, and spectral features.
- Before applying machine learning models, preprocessing is necessary to clean and prepare the data. This stage helps to ensure that irrelevant information is removed and that the dataset is structured properly for accurate classification

Dataset Description

- The dataset used consists of 100,000 observations categorized into three classes: Galaxy, Star, and Quasar. It includes both feature columns and identifier columns.
- The key features retained for classification include photometric values (u, g, r, i. z), positional data (alpha, delta), and spectral information (redshift).

Preprocessing Steps: 3.1 Columns Dropped vs Retained

- The dataset originally contained several identifier columns that do not contribute to classification and were therefore removed.
- Dropped Columns:
  - Obj_ID, run_ID, rerun_ID
  - Cam_col, field_ID
  - Spec_obj_ID, plate, MJD, fiber_ID
- Retained Columns:
  - U, G, R, I, Z
  - Alpha, delta
  - Redshift
  - class

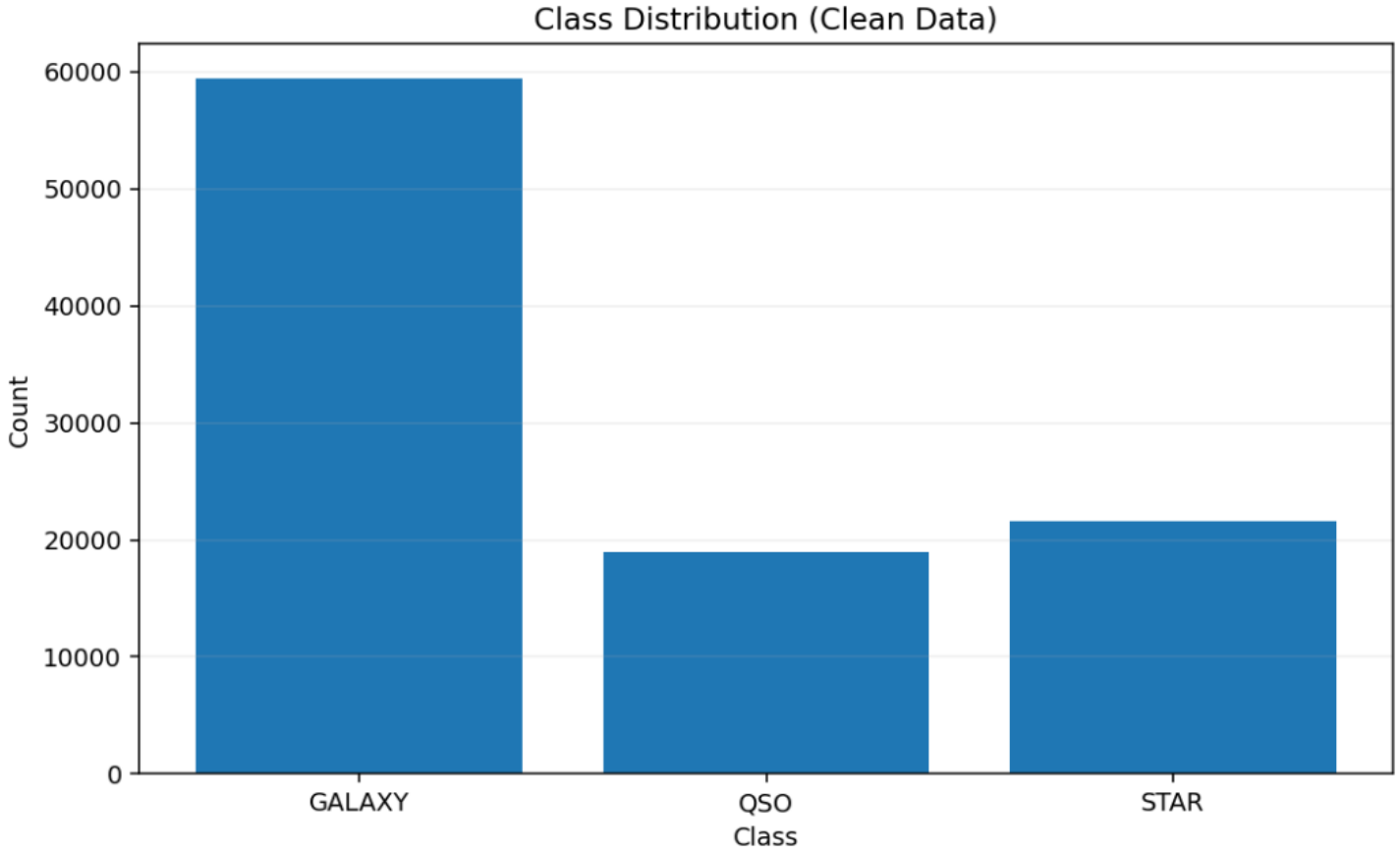
3.2 Data Splitting

- The dataset was split using a stratified sampling approach to maintain class balance across all subsets. The split ratios were 60% training, 20% validation, and 20% testing, using a fixed random seed of 4700 for reproducibility.

Feature Scaling

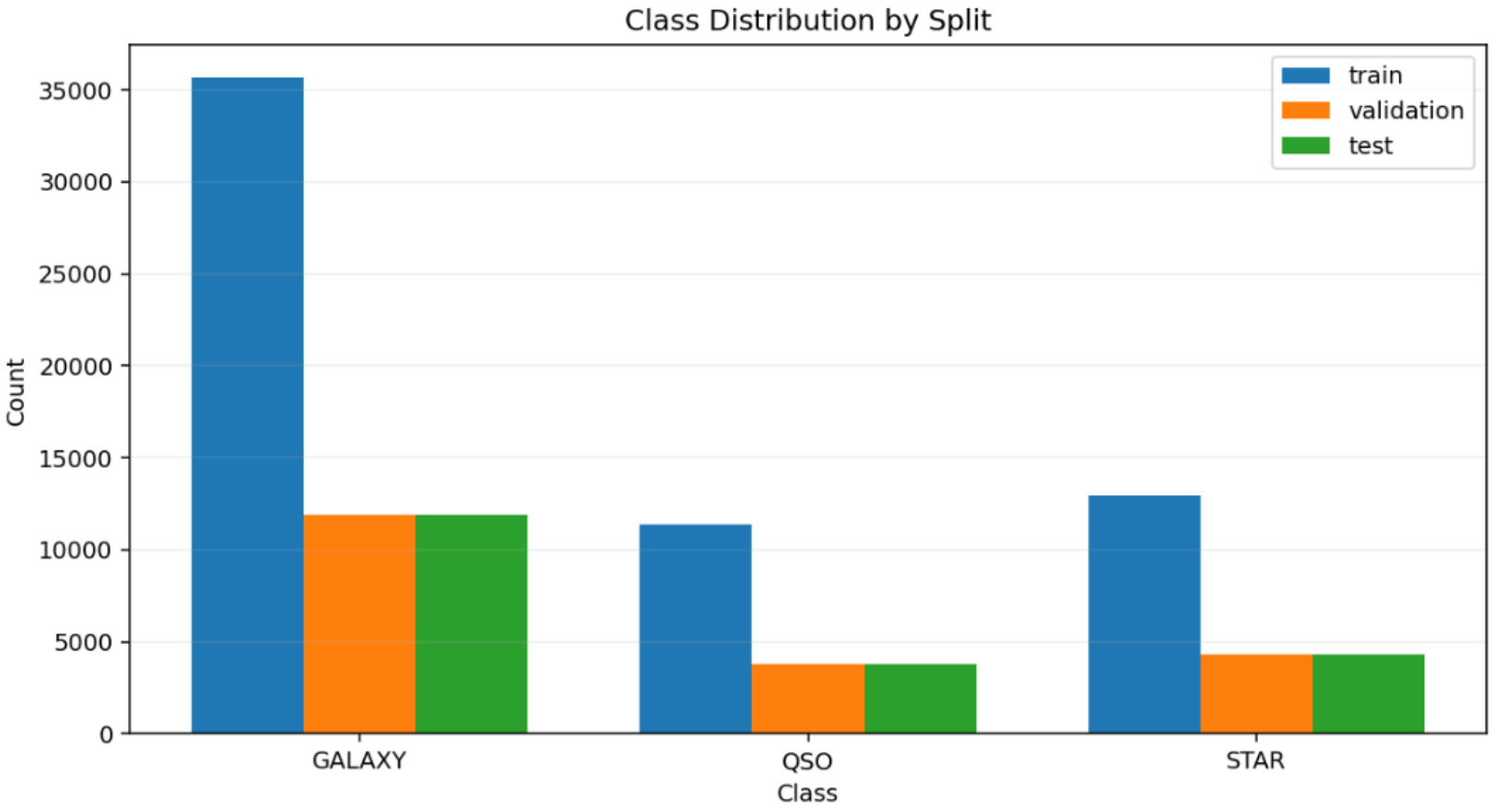
- Z-score normalization was applied to all feature columns using statistics calculated from the training set only. This ensures that all features are standardized and prevents data leakage.

Data Visualization and Analysis: 4.1 Clean Class Distribution



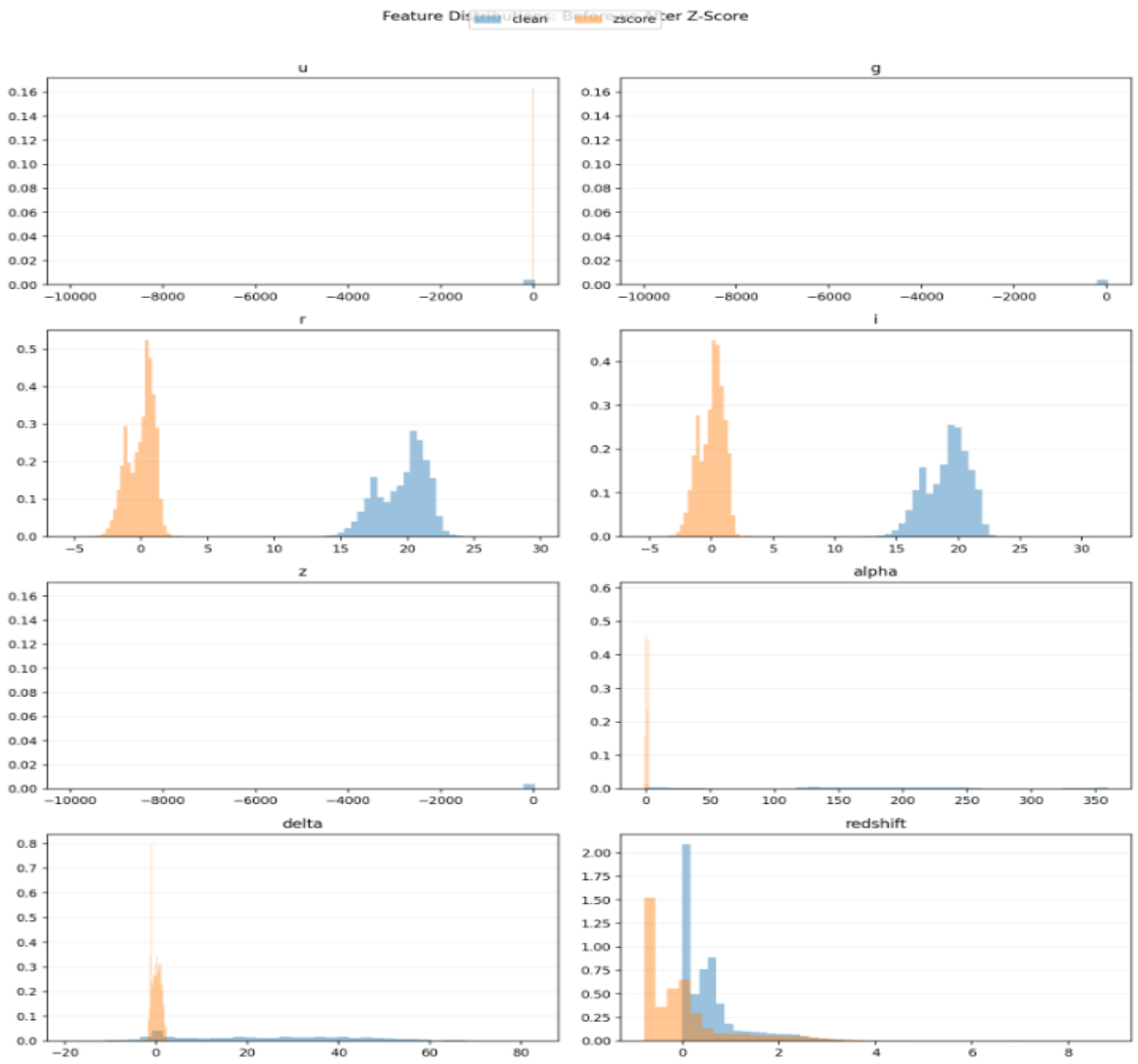
- The class distribution shows that galaxies are the most common class, followed by stars and quasars. This indicates a moderate imbalance in the dataset.

4.2 Split Distribution



- The class proportions are consistent across training, validation, and test sets, confirming that stratified sampling was successfully applied.

4.3 Before vs After Scaling



- Before normalization, feature values vary significantly in scale. After applying z-score normalization, all features are centered around zero with similar variance.

Raw vs Preprocessed Data Comparison

Raw

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	obj_ID	alpha	delta	u	g	r	i	z	run_ID	rerun_ID	cam_col	field_ID	spec_obj_	class	redshift	plate	MJD	fiber_ID
2	1.24E+18	135.6891	32.49463	23.87882	22.2753	20.39501	19.16573	18.79371	3606	301	2	79	6.54E+18	GALAXY	0.634794	5812	56354	171
3	1.24E+18	144.8261	31.27418	24.77759	22.83188	22.58444	21.16812	21.61427	4518	301	5	119	1.18E+19	GALAXY	0.779136	10445	58158	427
4	1.24E+18	142.1888	35.58244	25.26307	22.66389	20.60976	19.34857	18.94827	3606	301	2	120	5.15E+18	GALAXY	0.644195	4576	55592	299
5	1.24E+18	338.741	-0.40283	22.13682	23.77656	21.61162	20.50454	19.2501	4192	301	3	214	1.03E+19	GALAXY	0.932346	9149	58039	775
6	1.24E+18	345.2826	21.18387	19.43718	17.58028	16.49747	15.97711	15.54461	8102	301	3	137	6.89E+18	GALAXY	0.116123	6121	56187	842
7	1.24E+18	340.9951	20.58948	23.48827	23.33776	21.32195	20.25615	19.54544	8102	301	3	110	5.66E+18	QSO	1.424659	5026	55855	741
8	1.24E+18	23.23493	11.41819	21.46973	21.17624	20.92829	20.60826	20.42573	7773	301	2	462	1.25E+19	QSO	0.586455	11069	58456	113
9	1.24E+18	5.433176	12.06519	22.24979	22.02172	20.34126	19.48794	18.84999	7773	301	2	346	6.96E+18	GALAXY	0.477009	6183	56210	15
10	1.24E+18	200.2905	47.1994	24.40286	22.35669	20.61032	19.4649	18.95852	3716	301	5	108	7.46E+18	GALAXY	0.660012	6625	56386	719
11	1.24E+18	39.14969	28.10284	21.74669	20.03493	19.17553	18.81823	18.65422	5934	301	4	122	2.75E+18	STAR	-7.9E-06	2444	54082	232
12	1.24E+18	328.0921	18.22031	25.77163	22.52042	20.63884	19.78071	19.05765	8102	301	3	27	5.65E+18	GALAXY	0.459596	5020	55852	525
13	1.24E+18	243.9866	25.73828	23.76761	23.79969	20.98318	19.80745	19.45579	3927	301	4	112	5.32E+18	GALAXY	0.591409	4727	55693	855
14	1.24E+18	345.8019	32.67287	23.17274	20.14496	19.41948	19.22034	18.89359	8157	301	2	38	7.32E+18	STAR	7.18E-05	6504	56540	574
15	1.24E+18	331.502	10.0358	20.8294	18.75091	17.51118	17.01631	16.62772	7773	301	2	123	5.7E+18	GALAXY	0.152194	5065	55739	200
16	1.24E+18	344.9848	-0.35262	23.20911	22.79291	22.08589	21.86282	21.8512	4192	301	3	255	1.04E+19	GALAXY	0.81816	9215	57682	796
17	1.24E+18	244.8245	25.15456	24.8868	22.13311	20.44728	19.49171	18.9747	3927	301	4	118	5.32E+18	GALAXY	0.484929	4727	55693	23
18	1.24E+18	353.2015	3.080796	24.5489	21.44267	20.95315	20.7936	20.48442	7712	301	5	284	4.82E+18	STAR	-0.00043	4283	55864	178
19	1.24E+18	1.494389	3.291746	20.38562	20.40514	20.29996	20.05918	19.89044	7712	301	5	339	9.84E+18	QSO	2.031528	8743	57663	295
20	1.24E+18	14.38314	3.214326	21.82154	20.5573	19.94918	19.76057	19.55514	7712	301	5	425	9.86E+18	STAR	-0.00044	8753	57373	258
21	1.24E+18	167.1317	67.33994	20.48292	18.67807	17.6168	17.11936	16.73351	1412	301	5	124	5.52E+17	GALAXY	0.111588	490	51929	613
22	1.24E+18	171.9754	67.74745	22.13367	20.84772	18.96537	18.31696	17.98124	1412	301	5	136	5.53E+17	GALAXY	0.374756	491	51942	572
23	1.24E+18	144.7853	46.8265	24.54793	22.33601	20.92259	19.87177	19.16934	2821	301	2	28	8.24E+18	STAR	-0.00012	7316	56710	756
24	1.24E+18	145.273	46.96013	25.44243	20.77028	19.6617	19.08481	18.83176	2821	301	2	30	9.39E+17	GALAXY	0.66231	834	52316	115
25	1.24E+18	145.883	47.30048	21.73992	21.53095	21.26763	21.36257	21.15861	2821	301	2	33	8.23E+18	QSO	2.07568	7308	56709	596
26	1.24E+18	241.4263	27.22469	18.88323	17.54229	17.01789	16.75376	16.72259	3927	301	4	94	1.57E+18	GALAXY	0.032081	1393	52824	368
27	1.24E+18	132.9225	4.521865	21.2611	20.50495	18.36379	23.17828	17.96264	3015	301	3	126	1.34E+18	GALAXY	0.250956	1190	52670	190

Preprocessed

	A	B	C	D	E	F	G	H	I	J
1	split	u	g	r	i	z	alpha	delta	redshift	class
2	train	0.069983	0.057892	1.587586	1.187808	0.073624	-0.33892	0.361512	0.278156	GALAXY
3	train	0.081832	0.05379	0.521554	0.15109	0.008497	-0.36629	0.580537	0.093136	GALAXY
4	train	0.005531	0.080958	1.062409	0.809723	0.015871	1.67348	-1.2489	0.488223	GALAXY
5	train	-0.06036	-0.07034	-1.69847	-1.76985	-0.07465	1.741366	-0.15146	-0.63091	GALAXY
6	train	0.038515	0.070244	0.90603	0.668198	0.023085	1.696872	-0.18168	1.163239	QSO
7	train	0.060837	0.046289	0.521856	0.217371	0.008748	0.236672	1.171126	0.114824	GALAXY
8	train	-0.00399	-0.0104	-0.25272	-0.15108	0.001314	-1.43561	0.200286	-0.79014	STAR
9	train	0.094244	0.050287	0.537253	0.397309	0.011169	1.562967	-0.30213	-0.15997	GALAXY
10	train	0.045333	0.081523	0.723145	0.412544	0.020895	0.69014	0.080075	0.020762	GALAXY
11	train	0.030814	-0.00771	-0.12102	0.078029	0.007162	1.746755	0.432619	-0.79003	STAR
12	train	-0.02638	-0.04175	-1.15122	-1.17775	-0.04819	1.598355	-0.71822	-0.58145	GALAXY
13	train	0.031702	0.05694	1.318444	1.583624	0.079412	1.738276	-1.24635	0.331662	GALAXY
14	train	0.072648	0.04083	0.433839	0.232646	0.009143	0.698836	0.0504	-0.12523	GALAXY
15	train	0.064401	0.023972	0.706933	0.974419	0.046024	1.823547	-1.0718	-0.79071	STAR
16	train	-0.03721	-0.00136	0.354308	0.555972	0.031513	-1.82639	-1.06107	1.995324	QSO
17	train	-0.03484	-0.04353	-1.0942	-1.11904	-0.04561	-0.10744	2.19504	-0.63713	GALAXY
18	train	0.064378	0.045784	0.690435	0.449192	0.013898	-0.33935	1.152168	-0.79029	STAR
19	train	0.086209	0.007554	0.009743	0.000809	0.005651	-0.33429	1.158962	0.117974	GALAXY
20	train	0.099451	0.020844	-0.01759	-0.14336	-0.00789	-0.35489	-0.94616	-0.15766	GALAXY
21	train	-0.03361	-0.03956	-0.85834	-0.84292	-0.032	1.617133	-1.24754	-0.66524	GALAXY
22	train	-0.04284	-0.05098	-1.25833	-1.28793	-0.05269	1.656381	-1.24427	-0.58689	GALAXY
23	train	0.059803	-0.00203	-0.59909	-0.70842	-0.02822	0.781173	-0.2374	-0.26275	GALAXY
24	train	0.02856	0.038172	0.627089	0.466125	0.018933	0.650988	0.193242	-0.0414	GALAXY
25	train	0.023893	0.028621	0.225338	0.028055	0.008493	-0.34001	-0.93118	-0.09199	GALAXY
26	train	0.052634	0.062214	0.784429	0.595576	0.026449	-0.42779	-1.02791	-0.07498	GALAXY
27	train	-0.03035	0.042909	2.785062	1.341018	0.103237	-0.42618	0.609527	-0.7897	STAR

Raw vs Preprocessed Data: Explained

- The raw dataset contains original feature values along with identifier columns such as object IDS and observational metadata. These values vary in scale and include unnecessary information for classification.
- In contrast, the preprocessed dataset removes identifier columns and retains only relevant features. Additionally, z-score normalization is applied, resulting in values centered around zero. A split column is also added to indicate training, validation, and test data.
- This results in a cleaner and more consistent dataset suitable for machine learning.

Conclusion

- The preprocessing stage has successfully prepared the dataset for machine learning by removing irrelevant columns, applying stratified splitting, and normalizing features. These steps improve model performance and help to ensure reliable results.

Sources

- Fedesoriano. (January 2022). Stellar Classification Dataset - SDSS17. Retrieved from https://www.kaggle.com/fedesoriano/stellar-classification-dataset-sdss17.